

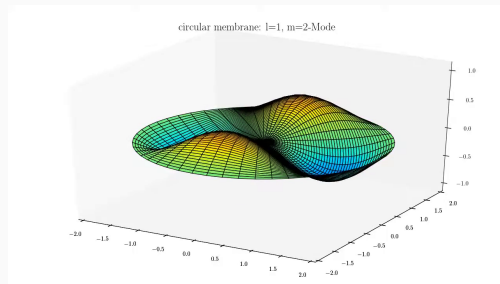
Analysis of free vibrations

Analysis of free and forced vibrations

ME473 Dynamic finite element analysis of structures

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Where do we stand?

Week	Module	Lecture topic	Mini-projects
1	Linear elastodynamics	Strong and weak forms	
2		Galerkin method	
3		Finite element method	Groups formation
4		Systematization of the procedure	Project 1 statement
5		3d elements, numerical integration	
6	Special structural elements	Bars and trusses	
7		Planar beams	Project 1 submission
8		Frames and grids	Project 2 statement
9		Kirchhoff-Love plates	
10		Reissner-Mindlin plates	
11		Shells	Project 2 submission
12	Free and forced vibrations	Analysis of free vibrations	Project 3 statement
13		Analysis of free vibrations	

Summary

- Recap week 12
- Numerical modal extraction algorithms for conservative systems
- Rayleigh's quotient
- Inverse iteration algorithm
- Subspace algorithm

Recommended readings

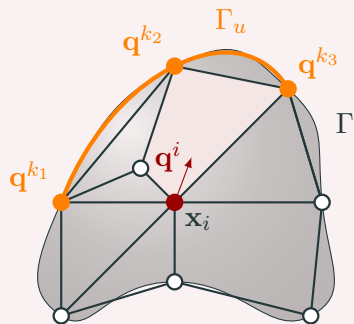
- (B) Bathe, Finite Element Procedures in Engineering Analysis (chap. 10 and 11)
- (P) Petyt, Introduction to finite element vibration analysis (chap. 11)
- (G) Gmür, Dynamique des structures (§4.1 and §4.2)
- (H) Hughes, The finite element method (chap. 10)

Recap week 12

Free vibrations of non-rotating conservative systems

The discretization of linear three-dimensional elastodynamics, as well as the analysis of vibrations in beams and plates via FEM, all lead to a system of ODE:

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{r}(t),$$



Free vibration: no external forcing is applied, i.e. $\mathbf{r}(t) = \mathbf{0}$.

- Generalized nodal displacements:

$$\mathbf{q}(t) = [\mathbf{q}^1(t), \dots, \mathbf{q}^n(t)]^T.$$

- **Boundary conditions:** $\mathbf{q}^k = \hat{\mathbf{q}}^k$ for all k such that $\mathbf{x}_k \in \Gamma_u$.
- **Initial conditions:** $\mathbf{q}(0) = \mathbf{u}_0$ and $\dot{\mathbf{q}}(0) = \mathbf{v}_0$

**Free undamped
discrete vibration
problem:**

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{0}$$



Ansatz:

$$\mathbf{q}(t) = \alpha \mathbf{p} \cos(\omega t + \varphi)$$



**Generalized
eigenvalue problem:**

$$(\mathbf{K} - \omega^2 \mathbf{M})\mathbf{p} = \mathbf{0}$$

Solving the eigenvalue problem:

$$n = \dim(\mathbf{K}) = \dim(\mathbf{M})$$

There are n eigenvalues λ_j and n corresponding eigenvectors $\mathbf{p}_j$ such that:

Eigenvalues (*natural frequencies squared*):
 $\lambda_j = \omega_j^2$ are the roots of the characteristic polynomial:

$$\det(\mathbf{K} - \lambda \mathbf{M}) = 0.$$

Eigenvectors (*modal shapes*): $\mathbf{p}_j$ are the solution of the equation

$$(\mathbf{K} - \lambda_j \mathbf{M})\mathbf{p}_j = \mathbf{0}.$$

Rigid body modes

In the semi-discrete weak form obtained via finite element discretization:

- The **mass matrix** $\mathbf{M}$ is symmetric and strictly positive definite.
- The **stiffness matrix** $\mathbf{K}$ is symmetric and positive semi-definite:

$$\mathbf{K}\mathbf{p} = \mathbf{0} \quad \text{for certain nonzero vectors } \mathbf{p}.$$

Consequently, the eigenvalues λ_j of the generalized eigenvalue problem are all real and non-negative:

$$0 \leq \omega_1 \leq \omega_2 \leq \cdots \leq \omega_n.$$

Rigid body modes: zero eigenvalues (i.e., $\omega_j = 0$) correspond to *rigid body motions*, where the system undergoes displacement without internal deformation.

Orthonormalization of mode shapes

Let $\mathbf{p}_i$ and $\mathbf{p}_j$ two eigenvectors corresponding to the eigenvalues λ_i and λ_j , then

$$\mathbf{p}_i^T \mathbf{M} \mathbf{p}_j = \delta_{ij} \quad \text{and} \quad \mathbf{p}_i^T \mathbf{K} \mathbf{p}_j = \omega_i^2 \delta_{ij}$$

where δ_{ij} represent Kronecker symbol.

Consequences: if we organize the modal vectors $\mathbf{p}_i$ in a so-called modal matrix $\mathbf{P}$:

$$\mathbf{P} = [\mathbf{p}_1 \mid \mathbf{p}_2 \mid \dots \mid \mathbf{p}_n]$$

then

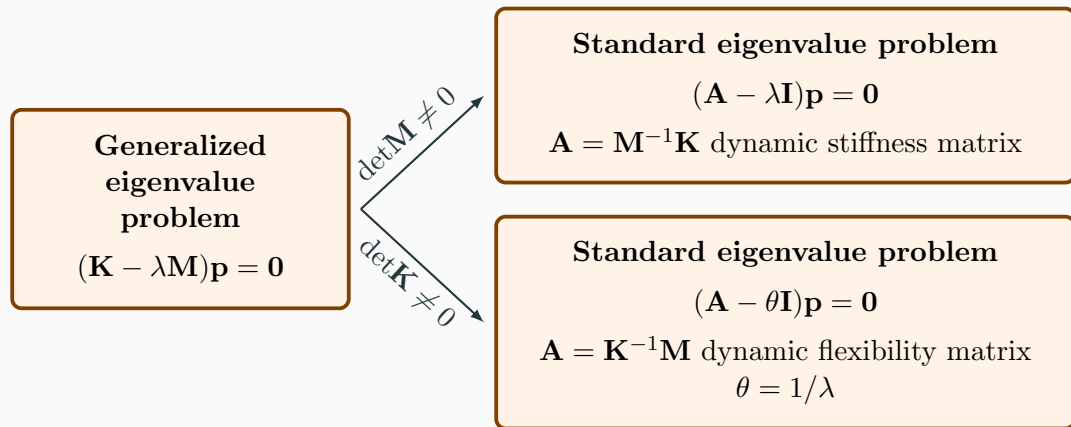
$$\mathbf{P}^T \mathbf{M} \mathbf{P} = \mathbf{I} \quad \text{and} \quad \mathbf{P}^T \mathbf{K} \mathbf{P} = \mathbf{\Lambda}$$

where $\mathbf{I}$ is the identity matrix of order n and $\mathbf{\Lambda}$ the *spectral matrix*:

$$\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n) = \text{diag}(\omega_1^2, \dots, \omega_n^2).$$

Numerical modal extraction algorithms

Transformation to standard form by inversion



The matrices $\mathbf{M}^{-1}\mathbf{K}$ and $\mathbf{K}^{-1}\mathbf{M}$ are not symmetric.

Transformation to standard form by decomposition

To reduce to a standard eigenvalue problem:

- Let $\mathbf{L}$ be the *Cholesky factor* of $\mathbf{M}$: $\mathbf{M} = \mathbf{L}\mathbf{L}^T$.

Since $\mathbf{M}$ is symmetric positive-definite, $\mathbf{L}$ is a real lower triangular matrix with positive diagonal entries.

- Define the change of variables $\mathbf{v} = \mathbf{L}^T \mathbf{p}$. Substituting into the original problem and multiplying by $\mathbf{L}^{-1}$ yields:

$$(\mathbf{K} - \omega^2 \mathbf{L}\mathbf{L}^T) \mathbf{p} = 0 \quad \Rightarrow \quad (\mathbf{L}^{-1} \mathbf{K} \mathbf{L}^{-T} - \omega^2 \mathbf{I}) \mathbf{v} = 0.$$

- Standard eigenvalue problem for symmetric and positive definite matrix $\mathbf{A} = \mathbf{L}^{-1} \mathbf{K} \mathbf{L}^{-T}$:

$$(\mathbf{A} - \lambda \mathbf{I}) \mathbf{v} = 0.$$

The eigenvalues or the standard problem are the same as those of the generalized problem, and the eigenvectors are related by $\mathbf{p} = \mathbf{L}^{-T} \mathbf{v}$.

Transformation to standard form by decomposition

- Cholesky decomposition can be applied to $\mathbf{K}$ as well, since $\mathbf{K}$ is symmetric positive-definite:

$$\mathbf{K} = \mathbf{L}\mathbf{L}^T.$$

- Standard eigenvalue problem for symmetric and positive definite matrix $\mathbf{A} = \mathbf{L}^{-1}\mathbf{M}\mathbf{L}^{-T}$:

$$(\mathbf{A} - \theta\mathbf{I})\mathbf{v} = \mathbf{0}.$$

The eigenvalues of the standard problem are related to those of the generalized problem by $\theta = 1/\lambda$, and the eigenvectors are related by $\mathbf{p} = \mathbf{L}^{-T}\mathbf{v}$.

Small and large scale problems

Economical computation algorithms for extracting $(\lambda_j, \mathbf{p}_j)$ for $1 \leq j \leq n_{modes}$ are required in practical applications, where $n_{modes} \ll n$ is the number of desired eigenpairs.

Small-scale problems

- system matrices are of modest size $n \leq 250$ or $250 \leq n \leq 2500$ and small bandwidth matrices.
- An explicit reduction to standard eigenvalue form is typically employed:
 - *Generalized Jacobi method*
 - *Householder QR-algorithm*

Large-scale problems

- $n \geq 2500$.
- The most effective algorithms are:
 - *Subspace iteration method,*
 - *Lanczos' method,*
 - *Irons-Guyan reduction method,*
 - *Arnoldi method*
 - *Davidson method.*

Rayleigh's quotient

Definition and first properties

Let $\mathbf{w}$ be a vector in $\mathbb{R}^n$, the **Rayleigh's quotient** associated to the matrix pair $(\mathbf{K}, \mathbf{M})$ is defined as the scalar function:

$$\mathcal{R}(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{K} \mathbf{w}}{\mathbf{w}^T \mathbf{M} \mathbf{w}}$$

Properties

- ① Homogeneity. Let α a non-zero constant, then

$$\mathcal{R}(\alpha \mathbf{w}) = \frac{(\alpha \mathbf{w})^T \mathbf{K} (\alpha \mathbf{w})}{(\alpha \mathbf{w})^T \mathbf{M} (\alpha \mathbf{w})} = \frac{\alpha^2 \mathbf{w}^T \mathbf{K} \mathbf{w}}{\alpha^2 \mathbf{w}^T \mathbf{M} \mathbf{w}} = \mathcal{R}(\mathbf{w})$$

- ② Rayleigh quotient of an eigenvector:

$$\mathcal{R}(\mathbf{w} = \mathbf{p}_i) = \frac{\mathbf{p}_i^T \mathbf{K} \mathbf{p}_i}{\mathbf{p}_i^T \mathbf{M} \mathbf{p}_i} = \lambda_i = \omega_i^2$$

Stationarity of Rayleigh's quotient

First variation of Rayleigh's quotient:

$$\begin{aligned}\delta\mathcal{R}(\mathbf{w}) &= \frac{2(\delta\mathbf{w}^T\mathbf{K}\mathbf{w})(\mathbf{w}^T\mathbf{M}\mathbf{w}) - 2(\mathbf{w}^T\mathbf{K}\mathbf{w})(\delta\mathbf{w}^T\mathbf{M}\mathbf{w})}{(\mathbf{w}^T\mathbf{M}\mathbf{w})^2} \\ &= \frac{2\delta\mathbf{w}^T}{\mathbf{w}^T\mathbf{M}\mathbf{w}} \left(\mathbf{K}\mathbf{w} - \frac{\mathbf{w}^T\mathbf{K}\mathbf{w}}{\mathbf{w}^T\mathbf{M}\mathbf{w}}\mathbf{M}\mathbf{w} \right) \\ &= \frac{2\delta\mathbf{w}^T}{\mathbf{w}^T\mathbf{M}\mathbf{w}} (\mathbf{K} - \mathcal{R}(\mathbf{w})\mathbf{M})\mathbf{w}\end{aligned}$$

Stationary Rayleigh's quotient in the vicinity of an modal shape !

If $\mathbf{w} = \mathbf{p}_i$ then

$$(\mathbf{K} - \mathcal{R}(\mathbf{w})\mathbf{M})\mathbf{w} = (\mathbf{K} - \lambda_i\mathbf{M})\mathbf{p}_i = 0$$

Rayleigh principle

Modal expansion:

$$\mathbf{w} = z_1 \mathbf{p}_1 + z_2 \mathbf{p}_2 + \cdots + z_i \mathbf{p}_i + \cdots + z_n \mathbf{p}_n = \mathbf{P} \mathbf{z}$$

where

$$\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_i, \dots, \mathbf{p}_n] \quad \text{modal matrix}$$

$$\mathbf{z} = [z_1, z_2, \dots, z_i, \dots, z_n]^T \quad \text{coefficient vector}$$

Then

$$\mathcal{R}(\mathbf{w} = \mathbf{P} \mathbf{z}) = \frac{\mathbf{z}^T \mathbf{P}^T \mathbf{K} \mathbf{P} \mathbf{z}}{\mathbf{z}^T \mathbf{P}^T \mathbf{M} \mathbf{P} \mathbf{z}} = \frac{\mathbf{z}^T \boldsymbol{\Lambda} \mathbf{z}}{\mathbf{z}^T \mathbf{z}} = \frac{\sum_{i=1}^n \lambda_i z_i^2}{\sum_{i=1}^n z_i^2}$$

If, without loss of generality, we impose $\mathbf{w}^T \mathbf{M} \mathbf{w} = 1$, then $\mathbf{z}^T \mathbf{z} = \sum_{i=1}^n z_i^2 = 1$ and

$$\mathcal{R}(\mathbf{w} = \mathbf{P} \mathbf{z}) = \sum_{i=1}^n \lambda_i z_i^2$$

Rayleigh principle (continued)

Let us consider an eigenvalue λ_j , then we can rewrite the Rayleigh's quotient as:

$$\begin{aligned}\mathcal{R}(\mathbf{w}) &= \sum_{i=1}^n \lambda_i z_i^2 = \lambda_j - \lambda_j + \sum_{i=1}^n \lambda_i z_i^2 = \lambda_j - \lambda_j \left(\sum_{i=1}^n z_i^2 \right) + \sum_{i=1}^n \lambda_i z_i^2 \\ &= \lambda_j + \sum_{i=1}^n (\lambda_i - \lambda_j) z_i^2\end{aligned}$$

With eigenvalues ordered as $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_i \leq \dots \leq \lambda_n$, we have:

- If $j = 1$, then $\lambda_i - \lambda_1 \geq 0$ for all i , hence

$$\mathcal{R}(\mathbf{w}) = \lambda_1 + \sum_{i=1}^n (\lambda_i - \lambda_1) z_i^2 \geq \lambda_1.$$

- If $j = n$, then $\lambda_i - \lambda_n \leq 0$ for all i , hence

$$\mathcal{R}(\mathbf{w}) = \lambda_n + \sum_{i=1}^n (\lambda_i - \lambda_n) z_i^2 \leq \lambda_n.$$

Rayleigh quotient bounding theorem

- Courant-Fischer formulas:

$$\mathcal{R}(\mathbf{w}) \geq \lambda_1 = \min_{\mathbf{w}} \mathcal{R}(\mathbf{w})$$

$$\mathcal{R}(\mathbf{w}) \leq \lambda_n = \max_{\mathbf{w}} \mathcal{R}(\mathbf{w})$$

- Bounding theorem:

$$\lambda_1 \leq \mathcal{R}(\mathbf{w}) \leq \lambda_n$$

Modal frequencies and shapes search via Rayleigh quotient minimization

(1) Find λ_1 by minimization:

$$\lambda_1 = \min_{\mathbf{w}} \mathcal{R}(\mathbf{w})$$

(2) Find $\mathbf{p}_1$ by solving $(\mathbf{K} - \lambda_1 \mathbf{M})\mathbf{p}_1 = 0$.

(3) Find λ_2 by minimization:

$$\lambda_2 = \min_{\mathbf{w}} [\mathcal{R}(\mathbf{w}); \mathbf{w}^T \mathbf{M} \mathbf{p}_1 = 0]$$

(4) Find $\mathbf{p}_2$ by solving $(\mathbf{K} - \lambda_2 \mathbf{M})\mathbf{p}_2 = 0$.

$\vdots$

(j) Find λ_j by minimization:

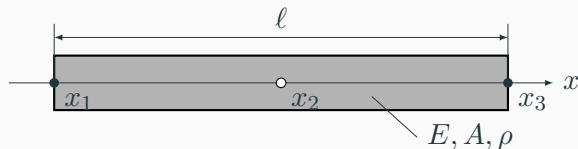
$$\lambda_j = \min_{\mathbf{w}} [\mathcal{R}(\mathbf{w}); \mathbf{w}^T \mathbf{M} \mathbf{p}_i = 0, i = 1, 2, \dots, j-1]$$

(j+1) Find $\mathbf{p}_j$ by solving $(\mathbf{K} - \lambda_j \mathbf{M})\mathbf{p}_j = 0$.

Example: Longitudinal vibrations of a free-free bar

Rayleigh quotient for longitudinal vibrations of free-free bar

Consider a free-free bar discretized by one bilinear element.



- ℓ length
- A cross-sectional area
- E Young's modulus
- ρ material density

Objective:

Compute the approximated natural frequencies and corresponding mode shapes via the minimization of Rayleigh's quotient

MATLAB example - longitudinal vibrations of free-free bar

► [Go to Matlab Drive](#)

Rayleigh-Ritz analysis

Rayleigh-Ritz analysis: fundamentals

Goal: approximate the lowest n_{modes} eigenvalues and corresponding eigenvectors of the generalized eigenproblem $\mathbf{K}\mathbf{p} = \lambda\mathbf{M}\mathbf{p}$

Inputs:

- $\mathbf{K}$, $\mathbf{M}$: positive definite stiffness and mass matrices (or shifted to be so). If $\mathbf{K}$ is singular, use shift: set $\mathbf{K}_\sigma = \mathbf{K} + \sigma\mathbf{M}$ with $\sigma > 0$.
- n_{modes} linearly independent **Ritz basis vectors** ψ_i , spanning a subspace $\mathcal{S}$.

Output:

- Approximated modal matrix of size $n \times n_{modes}$

$$\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_{n_{modes}}]$$

- Approximated spectral matrix:

$$\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_{n_{modes}})$$

Rayleigh-Ritz reduction algorithm

- ① Any trial vector in $\mathcal{S}$ is written as $\mathbf{w} = \sum_{i=1}^{n_{modes}} x_i \boldsymbol{\psi}_i$, with coordinates x_i .
- ② Insert $\mathbf{w} = \boldsymbol{\Psi} \mathbf{x}$ into $\mathcal{R}(\mathbf{w})$ to obtain the projected stiffness and mass matrices into the Ritz basis:

$$\tilde{\mathbf{K}} = \boldsymbol{\Psi}^T \mathbf{K} \boldsymbol{\Psi}, \quad \tilde{\mathbf{M}} = \boldsymbol{\Psi}^T \mathbf{M} \boldsymbol{\Psi},$$

where $\boldsymbol{\Psi} = [\boldsymbol{\psi}_1, \boldsymbol{\psi}_2, \dots, \boldsymbol{\psi}_{n_{modes}}]$.

- ③ Solve the reduced ($n_{modes} \times n_{modes}$) generalized eigenvalue problem by minimization:

$$\tilde{\mathbf{K}} \mathbf{x} = \phi \tilde{\mathbf{M}} \mathbf{x}.$$

to obtain approximate eigenpairs $(\phi_i, \mathbf{x}_i)$ for $i = 1, 2, \dots, n_{modes}$.

- ④ Compute the modal and the spectral matrices:

$$\tilde{\boldsymbol{\Lambda}} = \text{diag}(\phi_1, \phi_2, \dots, \phi_{n_{modes}}),$$

and the approximate mode shapes

$$\mathbf{p}_i = \boldsymbol{\Psi} \mathbf{x}_i.$$

Upper-bound property:

$$\lambda_i \leq \phi_i, \quad i = 1, \dots, n_{modes}.$$

- Ritz basis often obtained from static solutions:

$$\mathbf{K}\Psi = \mathbf{R},$$

where $\mathbf{R}$ contains chosen load patterns.

- Accuracy depends on how well $\mathcal{S}$ approximates the subspace of the lowest eigenmodes.
- Many well-known methods are special cases of Rayleigh–Ritz:
 - **Static condensation** (Irons-Guyan reduction)
 - **Component mode synthesis**
 - **Subspace iteration**

Inverse iteration algorithm

General idea of inverse and forward iteration

Goal: compute **one** eigenvector of the generalized eigenproblem $\mathbf{K}\mathbf{p} = \lambda\mathbf{M}\mathbf{p}$.

Idea: Use an *iterative* improvement of an assumed eigenvector.

- Start from an arbitrary vector $\mathbf{p}^{(0)}$.
- Generate a sequence of vectors $\mathbf{p}^{(1)}, \mathbf{p}^{(2)}, \dots$ that converge to an eigenvector.
- Compute an approximate eigenvalue using Rayleigh's quotient:

$$\lambda^{(k)} = \mathcal{R}(\mathbf{p}^{(k)}).$$

Inverse iteration:

- Form a “force” vector: $\mathbf{R}_1 = \mathbf{M}\mathbf{p}^{(0)}$.
- Solve the static-like equilibrium problem:
$$\mathbf{K}\mathbf{p}^{(1)} = \mathbf{R}_1,$$
- Repeat until convergence is met.

Forward iteration:

- Compute $\mathbf{R}_1 = \mathbf{K}\mathbf{p}^{(0)}$.
- Solve
$$\mathbf{M}\mathbf{p}^{(1)} = \mathbf{R}_1,$$
- Repeat until convergence is met.

Inverse iteration algorithm

Inputs:

- $\mathbf{K}$, $\mathbf{M}$: stiffness and mass matrices
- σ : spectral shift
- $\mathbf{p}^{(0)}$: initial guess vector (non-zero)
- ε : convergence tolerance

Algorithm:

① If $\mathbf{K}$ is singular, use shift: set $\mathbf{K}_\sigma = \mathbf{K} + \sigma\mathbf{M}$

② For $k = 1, 2, \dots$ until convergence:

- Solve for $\bar{\mathbf{p}}^{(k)}$ the static equilibrium problem

$$\mathbf{K}_\sigma \bar{\mathbf{p}}^{(k)} = \mathbf{M} \mathbf{p}^{(k-1)}$$

- $\mathbf{M}$ -normalize the vector $\bar{\mathbf{p}}^{(k)}$:

$$\mathbf{p}^{(k)} = \bar{\mathbf{p}}^{(k)} / \sqrt{(\bar{\mathbf{p}}^{(k)})^T \mathbf{M} \bar{\mathbf{p}}^{(k)}}$$

- Check convergence: $\|\mathbf{p}^{(k)} - \mathbf{p}^{(k-1)}\| < \varepsilon$

Output: approximated eigenvector $\mathbf{p}^{(k)}$ and eigenvalue $\lambda^{(k)} = \mathcal{R}(\mathbf{p}^{(k)})$.

Convergence of inverse iteration

- The inverse iteration converges to $\mathbf{p}_j$, the eigenvector associated with the eigenvalue closest to σ , if the initial guess $\mathbf{p}^{(0)}$ is not $\mathbf{M}$ -orthogonal to $\mathbf{p}_j$.
- If no shift is used, i.e. $\sigma = 0$, the method converges to the eigenvector associated with the *smallest eigenvalue* provided the orthogonality condition on the initial guess is met.

- The rate of convergence depends on the ratio

$$\rho = \frac{|\lambda_j - \sigma|}{|\lambda_{j+1} - \sigma|},$$

where λ_j is the eigenvalue closest to σ and λ_{j+1} is the next closest eigenvalue.

- If $\varepsilon = 10^{-2s}$, then approximately $2s$ correct digits of the eigenvector are obtained and s correct digits of the eigenvalue.

Subspace iteration method

General ideas behind the subspace iteration method

Goal: compute the lowest $n_{modes} \ll n$ eigenpairs $(\mathbf{p}_i, \lambda_i)$ of the generalized eigenproblem

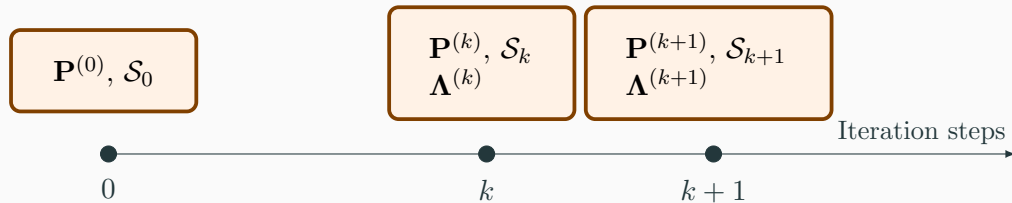
$$\mathbf{K}\mathbf{p} = \lambda\mathbf{M}\mathbf{p}.$$

Method developed and named by K. J. Bathe consists of the following three steps:

- ① Establish q starting iteration vectors, $q > n_{modes}$, where n_{modes} is the number of eigenvalues and vectors to be calculated.
- ② Use **inverse iteration** on the q vectors and **Rayleigh-Ritz analysis** to extract the "best" eigenvalue and eigenvector approximations from the q iteration vectors.
- ③ After iteration convergence, use the **Sturm sequence** check to verify that the required eigenvalues and corresponding eigenvectors have been calculated.

Preliminary considerations

Named the subspace iteration method because the iteration is equivalent to iterating with a q -dimensional subspace $\mathcal{S}_k = \text{span}(\mathbf{p}_1^{(k)}, \dots, \mathbf{p}_q^{(k)})$ and should not be regarded as a simultaneous iteration with q individual iteration vectors.



The effectiveness of the algorithm lies in that it is much easier to establish a q -dimensional starting subspace which is close to $\text{span}(\mathbf{p}_1, \dots, \mathbf{p}_{n_{modes}})$ than to find n_{modes} vectors that are each close to a required eigenvector $\mathbf{p}_i$.

Preliminary considerations

- All characteristics of the Ritz analysis pertain also to the subspace iteration; i.e., the smallest eigenvalues are approximated best in the iteration, and all eigenvalue approximations are **upper bounds** on the actual eigenvalues sought.
- Because the iteration is performed on a subspace, **convergence of the subspace** is all that is required and not convergence of individual iteration vectors to eigenvectors.
- If the initial vectors are linear combinations of the required eigenvectors, the solution algorithm converges in one step.

Pseudo-code of the subspace iteration method

Inputs:

- $\mathbf{K}$, $\mathbf{M}$: stiffness and mass matrices of dimension $n \times n$.
- n_{modes} : number of desired eigenmodes.
- q : number of iteration vectors, with $q > n_{modes}$.
- $\mathbf{P}^{(0)} \in \mathbb{R}^{n \times q}$: initial guess matrix of linearly independent vectors

$$\mathbf{P}^{(0)} = [\mathbf{p}_1^{(0)}, \dots, \mathbf{p}_q^{(0)}]$$

- σ : spectral shift (optional)
- ε : convergence tolerance

Output:

- Approximated eigenvectors: $\mathbf{P}^{(k)} = [\mathbf{p}_1^{(k)}, \dots, \mathbf{p}_q^{(k)}]$
- Approximated eigenvalues: $\mathbf{\Lambda}^{(k)} = \text{diag}(\lambda_1^{(k)}, \dots, \lambda_q^{(k)})$

Pseudo-code of the subspace iteration method

Algorithm:

- ① If $\mathbf{K}$ is singular, use shift: set $\mathbf{K}_\sigma = \mathbf{K} + \sigma\mathbf{M}$
- ② For $k = 1, 2, \dots$ until convergence:
 - Do step 1: perform inverse iteration on q vectors to obtain

$$\overline{\mathbf{P}^{(k)}}.$$

- Do step 2a: compute the projected stiffness and mass matrices:

$$\mathbf{K}^{(k)} \quad \text{and} \quad \mathbf{M}^{(k)}.$$

- Do step 2b: solve the projected generalized eigenvalue problem to obtain the projected modal matrix and the spectral matrix:

$$\mathbf{Z}^{(k)} \quad \text{and} \quad \mathbf{\Lambda}^{(k)}.$$

- Do step 2c: calculate the updated modal matrix $\mathbf{P}^{(k)}$.
- Do step 3: check convergence

Subspace iteration steps

- ❶ **Step 1:** Inverse iteration on q vectors: find the $(n \times q)$ matrix $\overline{\mathbf{P}}^{(k)}$ such that

$$\mathbf{K}\overline{\mathbf{P}}^{(k)} = \mathbf{M}\mathbf{P}^{(k-1)}$$

- ❷ **Step 2a:** Compute projected stiffness and mass matrices $(q \times q)$:

$$\mathbf{K}^{(k)} = (\overline{\mathbf{P}}^{(k)})^T \mathbf{K} \overline{\mathbf{P}}^{(k)}, \quad \mathbf{M}^{(k)} = (\overline{\mathbf{P}}^{(k)})^T \mathbf{M} \overline{\mathbf{P}}^{(k)}$$

- ❸ **Step 2b:** Solve $(q \times q)$ generalized eigenvalue problem: Find the projected modal matrix $\mathbf{Z}^{(k)}$ and the spectral matrix $\mathbf{\Lambda}^{(k)}$ such that

$$\mathbf{K}^{(k)} \mathbf{Z}^{(k)} = \mathbf{M}^{(k)} \mathbf{Z}^{(k)} \mathbf{\Lambda}^{(k)}$$

- ❹ **Step 2c:** Calculate the updated modal matrix $(n \times q)$:

$$\mathbf{P}^{(k)} = \overline{\mathbf{P}}^{(k)} \mathbf{Z}^{(k)}$$

Step 1 - Inverse iteration on q vectors

Suppose that the Step 1 is replaced by a simultaneous inverse iteration on m eigenvectors:

$$\mathbf{P}^{(k)} = (\mathbf{K}^{-1}\mathbf{M})\mathbf{P}^{(k-1)} = \dots = (\mathbf{K}^{-1}\mathbf{M})^k \mathbf{P}^{(0)}$$

Define the subspace $\mathcal{S}^{(k)}$ of rank q , spanned by the vectors $\{\mathbf{p}_i^{(k)}\}$.

$\mathbf{P}^{(k)} = [\mathbf{p}_1^{(k)}, \dots, \mathbf{p}_q^{(k)}]$ forms a *non-orthogonal* basis of $\mathcal{S}^{(k)}$.

✗ All columns of $\mathbf{P}^{(k)}$ tend toward $\mathbf{p}_1$

✗ Collinearity if no orthogonalization is applied !

Orthogonalization of vectors $\mathbf{p}_i^{(k)}$ at each iteration needed !

Modified Step 2 - Gram-Schmidt orthogonalization

- ① **Step 1:** Inverse iteration on q vectors: find the $(n \times q)$ matrix $\overline{\mathbf{P}}^{(k)}$ such that

$$\mathbf{K}\overline{\mathbf{P}}^{(k)} = \mathbf{M}\mathbf{P}^{(k-1)}$$

- ② **Step 2:** Grand-Schmidt orthogonalization:

$$\mathbf{P}^{(k+1)} = \mathbf{P}^{(k)}\mathbf{R}^k$$

where $\mathbf{R}^k$ is an upper triangular matrix chosen in such way that

$$(\mathbf{P}^{(k+1)})^T \mathbf{M} \mathbf{P}^{(k+1)} = \mathbf{I}$$

Gram-Schmidt method is inefficient as iteration vector $\mathbf{p}_i^{(k)}$ is forced to converge to a prescribed eigenvector $\mathbf{p}_i$ without allowing a more effective linear combination of the iteration vectors to take place.

Step 2 - Orthogonalization by minimization of the Rayleigh quotient

- Orthogonalization by minimization of the Rayleigh quotient:

$$\mathcal{R}(\mathbf{w}^{(k)}) = \frac{(\mathbf{w}^{(k)})^T \mathbf{K} \mathbf{w}^{(k)}}{(\mathbf{w}^{(k)})^T \mathbf{M} \mathbf{w}^{(k)}}$$

- Let $\mathbf{w}^{(k)} = \overline{\mathbf{P}^{(k)}} \mathbf{z}^{(k)}$

- Projected Rayleigh's quotient:

$$\mathcal{R}(\mathbf{w}^{(k)}) = \frac{(\mathbf{z}^{(k)})^T \mathbf{K}^{(k)} \mathbf{z}^{(k)}}{(\mathbf{z}^{(k)})^T \mathbf{M}^{(k)} \mathbf{z}^{(k)}}$$

where

$$\mathbf{K}^{(k)} = (\overline{\mathbf{P}^{(k)}})^T \mathbf{K} \overline{\mathbf{P}^{(k)}}, \quad \mathbf{M}^{(k)} = (\overline{\mathbf{P}^{(k)}})^T \mathbf{M} \overline{\mathbf{P}^{(k)}}$$

Step 2b - Solve projected generalized eigenvalue problem

- Minimization of the projected Rayleigh's quotient (generalized eigenvalue problem of dimension $q \times q$)
- Stationary condition:

$$\delta \mathcal{R}(\mathbf{w}^{(k)}) = 0 \quad \Rightarrow \quad \mathbf{K}^{(k)} \mathbf{z}^{(k)} = \lambda^{(k)} \mathbf{M}^{(k)} \mathbf{z}^{(k)}$$

- Solve via transformation method (e.g., Jacobi method):

$$\mathbf{K}^{(k)} \mathbf{Z}^{(k)} = \mathbf{M}^{(k)} \mathbf{Z}^{(k)} \mathbf{\Lambda}^{(k)}$$

- Ritz vectors and values:

$$\mathbf{Z}^{(k)} = [\mathbf{z}_1^{(k)}, \dots, \mathbf{z}_q^{(k)}], \quad \text{and} \quad \mathbf{\Lambda}^{(k)} = \text{diag}(\lambda_1^{(k)}, \dots, \lambda_q^{(k)})$$

Step 2c - Update modal matrix and orthogonality check

- Update the modal matrix:

$$\mathbf{P}^{(k)} = \overline{\mathbf{P}^{(k)}} \mathbf{Z}^{(k)}$$

- Orthogonality check:

$$(\mathbf{P}^{(k)})^T \mathbf{M} \mathbf{P}^{(k)} = \mathbf{I}$$

Step 3 - Convergence to stop iteration

Convergence criterion based on relative change of eigenvalues:

$$\frac{|\lambda_i^{(k)} - \lambda_i^{(k-1)}|}{\lambda_i^{(k)}} < \varepsilon, \quad i = 1, 2, \dots, n_{modes}$$

Starting iteration vectors $\mathbf{P}^{(0)}$

- The starting iteration vectors in $\mathbf{P}^{(0)}$ should not be too close to each other, i.e., they should be linearly independent.
- Higher convergence rate can be obtained by increasing q . However, using more iteration vectors will also increase the computer effort for one iteration. In practice

$$q = \min(2n_{modes}, n_{modes} + 8)$$

Empirical guidelines

- The first column of $\mathbf{P}^{(0)}$ should be the diagonal of $\mathbf{M}$. This ensures that all mass degrees of freedom are excited.
- The other columns in $\mathbf{P}^{(0)}$, except for the last column, should be unit vectors $\mathbf{e}_i$, with entries +1 at the degrees of freedom with the smallest ratios k_{ii}/m_{ii} , and the last column in $\mathbf{P}^{(0)}$ should be a random vector with values in $[0, 1]$.

Convergence of the subspace iteration method

- Iteration is performed with q vectors, $q > n_{modes}$, but convergence is measured only on the approximations obtained for the n_{modes} smallest eigenvalues.
- If the starting iteration vectors in $\mathbf{P}^{(0)}$, are not orthogonal to any one of the required eigenvectors $\mathbf{p}_1, \dots, \mathbf{p}_{n_{modes}}$, then as $k \rightarrow \infty$

$$\tilde{\mathcal{S}}_k = \text{span}(\mathbf{p}_1^{(k)}, \dots, \mathbf{p}_{n_{modes}}^{(k)}) \rightarrow \mathcal{S}_\infty = \text{span}(\mathbf{p}_1, \dots, \mathbf{p}_{n_{modes}})$$

- Assuming that in the iteration the vectors in $\mathbf{P}^{(k+1)}$ are ordered in such way that the i -th diagonal element in $\mathbf{\Lambda}^{(k+1)}$ is larger than the $(i-1)$ -st element for $i = 2, \dots, p$, then the i -th column in $\mathbf{P}^{(k+1)}$ converges to $\mathbf{p}_i$ and the convergence rate is

$$\lambda_i / \lambda_{p+1}$$

Although this is an asymptotic convergence rate, it indicates that the smallest eigenvalues converge fastest.

Final Sturm sequence check

After convergence is achieved, a final check based on the Sturm sequence property can be performed to ensure that the number of computed eigenvalues below a certain value is correct.

- The Sturm sequence property states that the number of eigenvalues of the generalized eigenvalue problem $\mathbf{K}\mathbf{p} = \lambda\mathbf{M}\mathbf{p}$ that are less than a given value λ^* is equal to the number of negative pivots obtained when performing an $\mathbf{LDL}^T$ factorization of the matrix $\mathbf{K} - \lambda^*\mathbf{M}$.
- This property can be used to verify that the number of computed eigenvalues below a certain threshold matches the expected count, providing an additional layer of validation for the results obtained from the subspace iteration method.